# SmartTask: A Context Aware Mobile Task Management System for Intelligent Task Prioritization and Adaptive Notifications

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***Abstract*—Task management applications are widely used to organize daily activities, but many existing systems still depend on static task lists, fixed reminders, and manual prioritization. These limitations can increase cognitive load, cause poorly timed notifications, and lead to missed tasks. To address this issue, the SmartTask application was developed as a context aware mobile task management system that identifies task relevance using user context. The system uses contextual signals such as location, time, user activity, device state, and connectivity to match tasks with the user's current situation. Based on this matching process, SmartTask dynamically prioritizes active tasks and delivers notifications only when they are suitable. The application was implemented as a native Android system using Java, Android Studio, Room SQLite, Android location services, activity recognition, WorkManager, and the Android Notification API. It supports task creation, editing, deletion, completion, snoozing, postponing, context acquisition, task matching, priority ordering, notification action handling, local data storage, and offline functionality. The system was evaluated with 8 participants over 12 to 15 days. During the evaluation, users created 211 tasks and completed 185 tasks, giving an 87.7 percent task completion rate. The ignored or missed notification rate was 16.5 percent, and the overall Likert mean score was 3.73 out of 5. These results show that SmartTask is a practical Minimum Viable Product (MVP) that demonstrates the feasibility of combining context awareness, dynamic task prioritization, and adaptive notifications for intelligent mobile task management.**

***Keywords— Context aware computing, mobile task management, adaptive notifications, task prioritization, Android application, productivity, user evaluation***

## I. INTRODUCTION

Individuals managing tasks in fast paced digital environments often experience cognitive overload due to constant multitasking, task switching, and digital interruptions. Previous studies show that interruptions can increase frustration, effort, and stress during task completion [1], while users frequently move between different work activities within short periods, creating continuous cognitive pressure [2]. In this situation, traditional to do list applications become less effective because they usually display all tasks in a similar way and require users to manually decide which task should be handled first. This manual decision making process increases mental effort and can reduce user engagement with task management tools [3].

Another major issue in daily task management is notification fatigue. Many task management applications depend on fixed reminders that are triggered at preselected times, without considering whether the user is busy, moving, in a meeting, or in a suitable situation to respond. As a result, reminders may arrive at inappropriate moments and users may ignore them over time [3]. Although these applications support basic task recording and reminders, they do not fully solve the problem of deciding which task is most relevant in the user's current situation.

Context aware computing provides a useful direction for improving task management because it allows a system to respond based on real time user and device conditions. Context can include information such as location, time, user activity, calendar state, device state, and connectivity [4]. Prior research has shown that context aware systems can improve usability and effectiveness in mobile applications [5][6]. Since modern smartphones can collect contextual information through sensors and system services, mobile task management systems can use this data to make task reminders and task ordering more relevant to the user's actual situation.

However, many existing task management applications and research prototypes do not provide a complete practical solution. Some systems support location based reminders [7] or early context aware task handling such as Castaway [8], but these approaches are often limited to one type of context. Other systems such as Taskminder [9], context based filtering approaches [10], and rule based assistants such as DoTT [11] either depend on manual rules or do not provide automatic adaptation in everyday mobile use. Therefore, a gap remains for a practical mobile task management system that combines multiple context signals, dynamic prioritization, and adaptive notification delivery in a single working application.

To address this gap, SmartTask was developed as a completed context aware Android task management application. The system uses contextual information such as location, time, user activity, device state, and connectivity to identify task relevance in real time. Based on this information, SmartTask matches active tasks with the user's current situation, dynamically updates task order, and delivers notifications only when the task is relevant and the interruption level is acceptable. The application was implemented as a native Android Minimum Viable Product (MVP) using Java, Android Studio, local Room SQLite storage, Android sensor services, context matching logic, task prioritization logic, and the Android notification system.

The main contribution of this paper is the design and implementation of a practical context aware mobile task management MVP that moves beyond static task lists and fixed reminders. The system demonstrates how multiple

contextual signals can be used to support real time task filtering, task relevance scoring, dynamic prioritization, and adaptive notification delivery. The paper also presents the architecture, implementation, and evaluation of SmartTask, showing how context awareness can support more relevant task management in daily routines. Although earlier design stages considered advanced behavioral learning and insight generation, these features were not fully implemented in the final system. They remain suitable future extensions that can enhance personalization and long term user adaptation. Previous research has shown that personalized task prioritization can reduce scheduling effort [12], and this concept can be further explored in future improvements of the system. Overall, SmartTask demonstrates the feasibility of combining context awareness, dynamic prioritization, and adaptive notifications to reduce cognitive effort, minimize unnecessary reminders, and improve intelligent mobile task management.

## II. Literature Review

### A. Existing Commercial Task Management Applications

Modern mobile task management applications mainly support task recording, due dates, reminders, and manual organization. Popular applications such as Todoist allow users to create tasks, set due dates, and use location based reminders, but they do not automatically adjust task order based on the user's changing context [13][14]. Any.do also provides reminder options such as task alerts and daily summaries, but these reminders remain fixed and do not adapt to user behavior or current user conditions [15]. Microsoft To Do includes My Day suggestions, which provide some level of personalization, but most task prioritization and reminder decisions still depend on user input [16]. Google Tasks and Google Keep also follow a simple reminder based approach, where users manually set task times without dynamic context based adjustment [17][18].

Other task management tools focus mainly on organizing work rather than making intelligent decisions. TickTick supports time and location reminders, but task priority is still handled as a manual setting [19]. Trello provides a visual board based workflow for managing tasks, but it does not automatically prioritize tasks based on urgency, location, activity, or user availability [20]. These tools are useful for general productivity, but they mostly depend on static task lists, fixed reminders, and manual prioritization. Therefore, commercial applications provide only limited support for reducing cognitive load and notification fatigue in real world task management.

### B. Context Aware Systems

Context aware systems have been studied as a way to make task reminders more relevant to the user's current situation. Early systems such as Castaway used time and location to support task reminders at suitable moments [8]. ContextGear also focused on location based reminders by using smartphone location data to trigger reminders when users reached specific areas. These systems showed that context can improve the usefulness of reminders, especially when tasks are connected to places or times.

In collaborative task settings, systems such as Overseer used context information to support task assignment based on user roles and conditions [21]. This showed that context awareness can be useful not only for individual task reminders, but also for assigning and managing tasks in changing environments. However, many of these systems focused on specific use cases or a limited set of context signals. They did not provide a complete mobile task management solution that combines multiple context inputs, dynamic task ordering, and adaptive notifications for everyday users.

### C. Adaptive Notification Research

Another important area of research is adaptive notification timing. Poorly timed notifications can interrupt users and reduce productivity, especially when users are busy, moving, or focused on another activity. Studies have shown that contextual information such as user activity, task phase, and interruption state can help identify better moments for sending notifications [22]. The OASIS framework introduced the idea of delaying notifications until suitable breakpoints, reducing interruption cost and improving user performance [23]. This supports the idea that notifications should not only be based on task deadlines, but also on the user's current readiness to respond.

Sensor based systems have also been used to improve reminder timing. For example, a context aware reminder system used motion sensors such as accelerometers to detect user activity and trigger reminders based on movement patterns [24]. The AWARE framework provided a broader smartphone context collection platform that gathered data such as location, activity, connectivity, and other device signals to generate useful context for applications [25]. In addition, task prediction research showed that contextual patterns and usage behavior can be used to predict future tasks and support better scheduling [26]. These studies show the value of context aware and adaptive approaches, but they are often implemented as research prototypes rather than complete practical task management applications.

### D. Summary of Limitations in Existing Systems

The reviewed literature shows that commercial task management applications are useful for basic task recording and reminders, but they mostly lack real time context awareness and dynamic prioritization. Their reminders are usually fixed, and task ordering depends heavily on manual user decisions. As a result, users still need to decide which task is most important at a given moment, which can increase mental effort and reduce the effectiveness of the application.

Research systems provide stronger evidence that context awareness, adaptive notification timing, and task prediction can improve task management. However, many of these systems focus on limited features such as location reminders, activity based triggers, adaptive interruption handling, or task prediction as separate research problems [8][22][23][24][25][26]. Other studies, such as MageMemo, improve engagement through gamification but do not use context awareness [3], while interruption studies suggest useful rules for notification management without providing a full task management implementation [27]. Therefore, no single existing system fully combines multi context awareness, dynamic task prioritization, and adaptive notification delivery in a practical mobile task management application.

Based on this review, SmartTask was developed to address the limitations of both commercial applications and research prototypes. The system combines multiple contextual inputs such as location, time, user activity, device state, and

connectivity with task relevance scoring, dynamic prioritization, and adaptive notifications.

## III. Research Gap and Problem Statement

### A. Identified Research Gap

Existing task management systems provide reminder and scheduling features, but many still rely on static prioritization and limited context awareness [13][15][20]. Several research studies demonstrate the effectiveness of context aware reminders and adaptive notifications; however, most systems focus on only a few contextual factors or operate in controlled environments [8][22][23][24]. There is still limited implementation of mobile task management systems that combine multiple contextual signals such as time, location, activity, device state, and connectivity to support real time task prioritization and notification delivery.

SmartTask addresses this gap by implementing a context aware Android application that dynamically prioritizes tasks and delivers adaptive notifications based on the user's current context.

### B. Problem Statement

Traditional task management applications often require manual prioritization and deliver reminders without considering the user's current situation, which can lead to unnecessary interruptions and reduced task relevance [1][2][16]. Although previous studies have explored context aware reminders and adaptive scheduling, there remains a lack of unified mobile systems that integrate multiple contextual factors for real time task management [22][24][26].

This study addresses the problem by developing SmartTask, a context aware mobile application that supports dynamic task prioritization and adaptive notification delivery using real time contextual information.

## IV. Objectives

The main objective of this research was to develop SmartTask as a context aware mobile task management system that dynamically prioritizes tasks and adapts notifications based on user context to improve task relevance and reduce unnecessary interruptions.

The specific objectives of the study were as follows:

- To integrate contextual signals such as location, time, user activity, calendar state, device state, and connectivity into the task management process.
- To develop dynamic task prioritization mechanisms based on task urgency and contextual relevance.
- To implement adaptive notification delivery that minimizes interruptions and improves reminder relevance.
- To evaluate the effectiveness of the system using task completion, notification response, and user feedback.
- To identify future enhancements for improving personalization and long term context adaptation.

## V. Methodology and System Implementation

### A. Overall Methodology

SmartTask was developed using a prototype based methodology to evaluate the effectiveness of context aware mobile task management. The system was designed as a modular Android MVP, where the main components were implemented and tested separately before integration into a complete application.

The methodology focused on three main stages: system design and implementation, integration of task management and notification components, and evaluation through user testing and feedback analysis. The implemented prototype was used to examine how contextual information could support task prioritization and notification delivery in a practical mobile environment. Advanced behavioral learning and detailed insight generation were identified as future enhancements beyond the scope of the current MVP.

### B. System Architecture

SmartTask was designed using a modular architecture consisting of the Task Input Module, Context Acquisition Engine, Context Matching Module, Task Prioritization Module, and Adaptive Notification System. These components operate together to support task creation, context processing, task prioritization, and notification delivery.

The main architectural components are described below.

- Task Input Module: Handles task creation and management with properties such as title, deadline, category, preferred time, and optional location.
- Context Acquisition Engine: Collects contextual information including time, location, activity, device state, and connectivity.
- Context Matching Module: Evaluates task relevance based on the current context.
- Task Prioritization Module: Dynamically ranks active tasks according to urgency and contextual relevance.
- Adaptive Notification System: Delivers reminders only when notification conditions are suitable.

The architecture also considered future support for behavioral learning and personalized insights through extended usage analysis. Fig. 1 shows the high level SmartTask architecture.

### C. Implementation

SmartTask was implemented as a native Android MVP using Java. The application used Room SQLite for local data storage, enabling offline functionality and on-device processing of task and context data.

Several Android technologies and frameworks were integrated to support the implemented features:

- Room SQLite: Local storage for tasks, context data, and usage records.
- WorkManager: Background processing for scheduled context evaluation tasks.
- Google Maps and Places: Location selection and geofence support.
- Android Notification API: Reminder notifications and notification actions.
- SpeechRecognizer: Voice based task creation support.

- OpenAI Features: Optional voice command parsing and description insight analysis.

The implementation primarily focused on validating the MVP functionality through local processing, background context handling, and adaptive notification support.

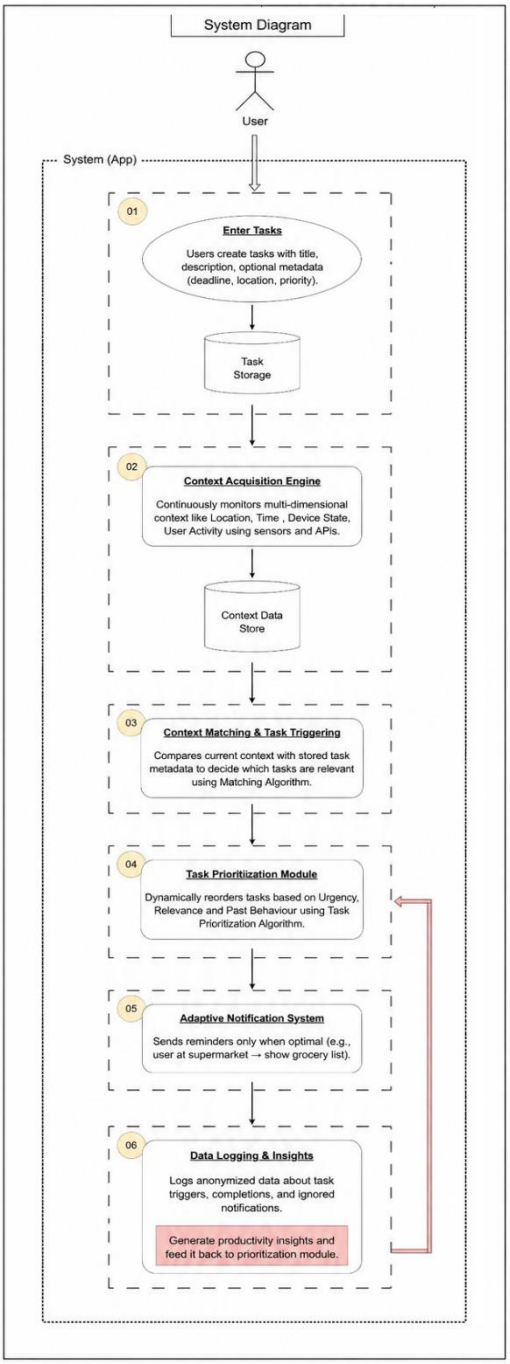


Fig. 1. High level system architecture of SmartTask

## D. Core System Logic

The core logic of SmartTask follows a structured processing pipeline. The system captures the user's context, compares it with stored task data, prioritizes active tasks, and decides whether a notification should be shown. Fig. 2 illustrates the overall SmartTask processing flow from task creation to context matching, prioritization, and notification delivery.

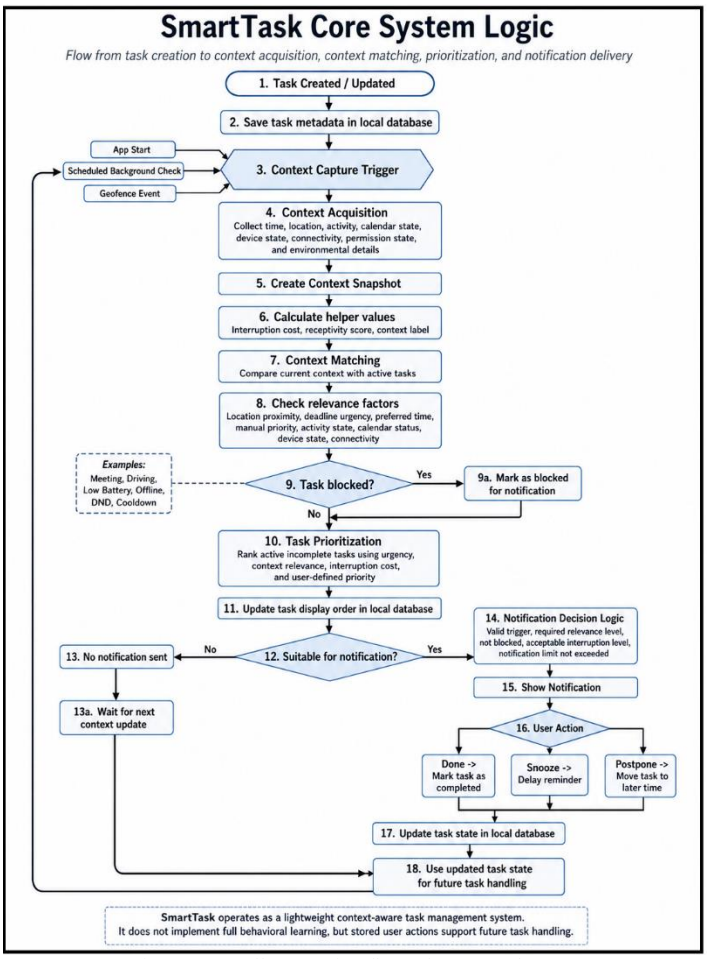


Fig. 2. Overall processing flow of SmartTask

- **Context Acquisition:** The system captures context when the app starts, during scheduled background checks, and when geofence events occur. Context collectors gather data such as time, location, activity, calendar state, device state, connectivity, permission state, and environmental details. These values are stored as context snapshots. The system also calculates helper values such as interruption cost, receptivity score, and context label.
- **Context Matching:** After a context snapshot is created, the system compares it with active tasks. It checks whether each task is relevant based on factors such as location proximity, deadline urgency, preferred time, manual priority, activity state, calendar status, device state, and connectivity. Tasks can be blocked from notification during unsuitable conditions such as meetings, driving, low battery, offline status, Do Not Disturb mode, or cooldown periods.
- **Task Prioritization:** The prioritization logic ranks active incomplete tasks using urgency, context relevance, interruption cost, and user defined priority. Tasks with higher relevance and urgency are placed higher in the task list. The updated order is saved using the display order value in the local database, allowing the task list to change based on the user's current situation.

- **Notification Decision Logic:** The notification system checks whether a task is suitable for alerting the user. A notification is sent only when the task has a valid trigger, meets the required relevance level, is not blocked, and has an acceptable interruption level. The system also limits the number of notifications in one processing cycle to reduce notification overload.
- **Notification Action Handling:** When a notification is shown, the user can mark the task as Done, Snooze it, or Postpone it. These actions update the task state in the local database and influence later task handling. This allows the system to continue adapting task order and notification behavior as the user interacts with the app.

This processing flow enabled SmartTask to function as a lightweight context aware task management system while supporting future expansion for behavioral learning and personalization.

### *E. Testing and Evaluation*

The SmartTask system was evaluated using both technical testing and user based evaluation to assess system functionality, integration, and practical usability.

The testing and evaluation approach included the following methods.

- Unit Testing: Verified task validation, context matching, prioritization, and notification logic.
- UI Testing: Evaluated user interactions such as task creation, editing, completion, and navigation.
- Integration Testing: Tested communication between task management, context processing, prioritization, and notification modules.
- Emulator Testing: Simulated different contextual and device conditions before deployment on physical devices.
- Real Device Testing: Validated sensor-based features including location tracking, activity recognition, and notification delivery.
- User Evaluation: Participants used SmartTask during daily activities and provided feedback through questionnaires.

The user evaluation involved students and working professionals over an extended testing period. The evaluation metrics included task completion rate, ignored or missed notification rate, user satisfaction, and system responsiveness. Table I summarizes the main evaluation results.

TABLE I. USER EVALUATION RESULTS SUMMARY

| Metric | Value |
|---|---|
| Participants | 8 |
| Total active testing days | 109 |
| Average active days per participant | 13.62 |
| Active days range | 12 to 15 |
| Total tasks created | 211 |
| Average tasks created per participant | 26.38 |
| Total tasks completed | 185 |
| Average tasks completed per participant | 23.12 |
| Overall task completion rate | 87.7% |
| Mean participant completion rate | 88.6% |
| Completion rate range | 81.6% to 96.2% |
| Total notifications received | 194 |
| Average notifications received per participant | 24.25 |
| Total notifications ignored or missed | 32 |
| Average ignored or missed notifications per participant | 4 |
| Overall ignored or missed notification rate | 16.5% |
| Mean participant ignored or missed rate | 16.2% |
| Ignored or missed rate range | 9.1% to 28.1% |
| Overall Likert mean | 3.73 |

## VI. RESULTS AND DISCUSSION

### *A. System Performance Results*

The user evaluation involved 8 participants, including students and working professionals, who used SmartTask over a testing period of 12 to 15 days. During the evaluation, participants created 211 tasks and completed 185 tasks, resulting in an overall task completion rate of 87.7%. Participants also received 194 notifications, of which 32 were ignored or missed, giving an ignored or missed notification rate of 16.5%.

These results indicate that participants actively used SmartTask for daily task management and generally responded to the delivered reminders. The findings support the practical feasibility of combining task prioritization and adaptive notification delivery within a context aware mobile task management MVP. However, the results should be interpreted as MVP validation due to the limited participant group and short evaluation period.

### *B. Likert Scale Results*

The questionnaire evaluated usability, context awareness, notification usefulness, productivity, satisfaction, and adoption using a five point Likert scale. The overall Likert mean score was 3.73 out of 5, indicating a generally positive user response. Fig. 3 shows the category level questionnaire results.

Usability and navigation received the highest mean score of 3.88, followed by notification usefulness and timing at 3.84. Context awareness and prioritization received a score of 3.71, while productivity, satisfaction, and adoption received 3.50.

These results suggest that users found the system easy to use and considered the adaptive reminders useful, while longer evaluation periods may be required to assess long term productivity impact.

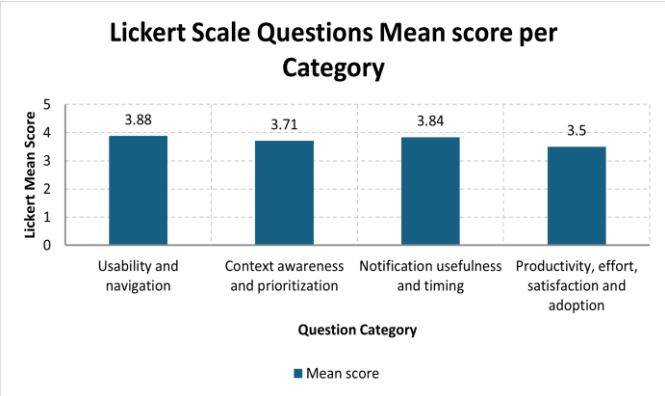


Fig. 3. Category level questionnaire results

### C. *Open Ended Feedback Summary*

The open ended feedback highlighted both positive aspects and improvement areas of the SmartTask MVP.

Positive Feedback:

- Context-aware and location-based reminders were consistently identified as the most useful feature.
- Participants reported reduced need to manually check task lists.
- Automatic task prioritization improved awareness of relevant tasks.

Challenges:

- Users reported issues related to notification timing, notification frequency, and location accuracy.
- Some participants found the task prioritization logic difficult to understand.

Participants also suggested improvements such as better notification controls, interface refinement, dashboard support, and stronger calendar and voice integration.

### D. *Interpretation of Findings*

The evaluation results demonstrate that SmartTask successfully functioned as a practical context aware task management MVP. The findings suggest that combining contextual information with task prioritization and adaptive notifications can improve reminder relevance and support daily task management activities. However, the study also identified limitations related to notification timing, prioritization transparency, and the short evaluation period. Therefore, the results should be considered as proof of concept validation rather than broad statistical evidence.

## VII. Social, Security and Ethical Considerations

SmartTask processes task and contextual information locally using Room SQLite, allowing core functions such as task prioritization and notification handling to operate on the user's device. Sensitive permissions including location, activity recognition, calendar access, notifications, and voice input are requested only when required, and related features remain disabled if permission is denied. User evaluation was conducted with participant consent, and collected feedback was used only for research purposes.

The system also considered notification ethics by reducing reminders during unsuitable situations such as meetings, driving, or Do Not Disturb mode. However, the MVP still has privacy limitations because some features depend on sensor access and contextual data collection. In addition, optional AI based features may involve external data processing, highlighting the need for clearer consent controls and improved privacy management in future versions.

## VIII. Limitations and Future Work

SmartTask was developed as a functional MVP to validate context aware task management, dynamic prioritization, and adaptive notifications. However, the system had several limitations, including limited behavioral learning, dependence on sensor permissions, and a small scale user evaluation with a short testing period. Furthermore, the evaluation did not compare SmartTask with an existing task management application under the same conditions. In addition, advanced analytics, personalization features, and the full Data Logging and Insights module were not fully implemented in the current version.

Future work can improve SmartTask through machine learning based personalization, improved notification timing, enhanced privacy controls, and larger scale user evaluations. Additional improvements such as clearer task prioritization explanations, better voice interaction, calendar integration, and interface enhancements can further improve long term usability and personalization.

## IX. Conclusion

SmartTask demonstrated the feasibility of a context aware mobile task management system that supports dynamic task prioritization and adaptive notification delivery. The developed MVP showed that contextual information can be used to improve task relevance and reduce unnecessary interruptions in everyday mobile task management.

The evaluation results indicated positive user acceptance and practical usability of the system. Participants actively used the application during the testing period and generally found the adaptive reminders and task prioritization features useful for managing daily activities.

Although the system was implemented as an MVP with a limited evaluation scope, the findings support the potential of context aware productivity systems in mobile environments. Future improvements can focus on behavioral learning, stronger personalization, improved notification accuracy, and larger scale evaluations to further enhance long term usability and effectiveness.

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